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2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

Mathematics

Calculus: Continuity & AOD

Continuity, Differentiability, Logarithmic/Parametric Derivatives & Applications of Derivatives · Section A Unit 2 & Section B1 Unit III

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment & Exam Blueprint

Sub-Topic / Domain	Official NTA CUET UG Syllabus Scope (Mathematics 319)	Exam Cognitive Focus
Continuity & Smoothness	Continuity and differentiability; Derivative of composite functions, chain rule; Derivatives of inverse trigonometric functions (like $\sin^{-1}x, \cos^{-1}x, \tan^{-1}x$); Derivative of implicit functions; Algebra of continuous functions; Relation between continuity and differentiability.	Determining piecewise continuity parameters (k); recognizing non-differentiable corner/cusp points ($\lvert x-c\rvert$); applying chain rule to nested inverse trigonometric functions; evaluating implicit $d\frac{y}{d}x$.
Advanced Differentiation	Concept of exponential and logarithmic functions; Derivatives of logarithmic and exponential functions; Logarithmic differentiation $(u(x)^{v(x)})$; Derivative of functions in parametric form $(x=f(t), y=g(t))$; Second-order derivatives $(d^2\frac{y}{d}x^2)$; Mean Value Theorems (Rolle's & Lagrange's MVT).	Executing logarithmic differentiation on variable towers; solving parametric second derivatives without missing the $d\frac{t}{d}x$ multiplier; computing intermediate c for Rolle's and LMVT.
Rate of Change & Geometry	Rate of change of quantities $(d\frac{y}{d}t)$; Related rates (spheres, cones, sliding ladders); Tangents and normals to curves; Slopes $(m=d\frac{y}{d}x)$ and equations of tangent and normal lines; Orthogonal intersection of curves $(m_1m_2=-1)$; Horizontal and vertical tangents.	Formulating geometric rate-of-change models; calculating rates of volume and surface area growth; finding points on curves with specified tangent slopes; testing curve orthogonality.
Monotonicity & Extrema	Increasing and decreasing functions; Strictly increasing/decreasing intervals on open intervals (a,b); Critical points $(f'(x)=0$ or undefined); First derivative test (sign changes); Second derivative test $(f''(c)<0$ for max, $f''(c)>0$ for min); Inconclusive $f''(c)=0$ cases.	Applying the Wavy Curve sign scheme to $f'(x)$; classifying critical points into local maxima, minima, or inflection points; quoting exact intervals of increase and decrease.
Optimization & Approximations	Maxima and minima in practical real-life word problems; Constructing single-variable objective functions; Absolute (global) extrema on closed intervals $[a,b]$; Checking boundary endpoints vs interior critical points; Linear approximations using differentials $(\Delta y \approx f'(x)\Delta x)$.	Setting up optimization word problems (box, inscribed cylinder/cone, wire cutting); testing boundary values $f(a), f(b)$ for absolute extrema; calculating percentage and relative errors.

EXAM PATTERN

CUET UG Scoring Blueprint: Calculus is the single largest thematic weight in the CUET UG Mathematics examination. It appears prominently as **Section A Unit 2 (compulsory for all candidates)** and constitutes the core of **Section B1 Unit III (Core Mathematics)** and **Section B2 Unit III (Applied Mathematics)**. Expect **8 to 12 questions (40 to 60 marks)** across the paper. Primary NTA traps include: **parametric second derivatives missing $d\frac{t}{d}x$**, **forgetting boundary endpoints in absolute extrema**, and **confusing non-differentiable corners with discontinuities**.

Continuity & Differentiability Foundations

1. Mathematical Definition of Continuity

A function $f(x)$ is defined to be **continuous at a point $x = c$** in its domain if and only if the following three criteria are simultaneously satisfied:

1. $f(c)$ is defined and finite (the point exists on the curve).
2. The limit $\lim_{x\to c} f(x)$ exists, meaning the left-hand limit equals the right-hand limit:
$$\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = L$$
3. The limit value strictly equals the function value:
$$\lim_{x\to c} f(x) = f(c)$$

Geometrically, a function is continuous on an interval if its graph can be drawn without lifting the pen from the paper.

Continuity in Intervals

- **Open Interval (a, b):** f is continuous on (a, b) if it is continuous at every single point $c \in (a, b)$.
- **Closed Interval $[a, b]$:** f is continuous on $[a, b]$ if:
  1. f is continuous on (a, b).
  2. f is right-continuous at a: $\lim_{x\to a^+} f(x) = f(a)$.
  3. f is left-continuous at b: $\lim_{x\to b^-} f(x) = f(b)$.

Algebra of Continuous Functions

Let f and g be two real functions continuous at $x = c$. Then:

1. $f + g$ and $f - g$ are continuous at c.
2. kf is continuous at c for any scalar $k \in \mathbb{R}$.
3. $f \cdot g$ is continuous at c.
4. $\frac{f}{g}$ is continuous at c, provided $g(c) \neq 0$.
5. **Composite Function Continuity:** If g is continuous at c and f is continuous at $g(c)$, then $(f \circ g)(x) = f(g(x))$ is continuous at c.

- **Universally Continuous Functions:** Polynomial functions, exponential functions (a^x), sine and cosine functions are continuous across the entire real line $\mathbb{R}$. Rational functions are continuous at all points where their denominators are non-zero.

2. Differentiability at a Point

Let f be a real function and c a point in its domain. The **derivative of f at c** is defined by:

$$f'(c) = \lim_{h\to 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists and is finite.

- **Left-Hand Derivative ($Lf'(c)$):** $Lf'(c) = \lim_{h\to 0} \frac{f(c-h)-f(c)}{-h}$.
- **Right-Hand Derivative ($Rf'(c)$):** $Rf'(c) = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$.

f is **differentiable at $x = c$** if and only if $Lf'(c) = Rf'(c)$.

The Fundamental Implication Theorem

THE GOLDEN RULE OF SMOOTHNESS

Theorem: Every differentiable function is necessarily continuous:

$$\text{Differentiability} \implies \text{Continuity}$$

The Converse is Strictly FALSE: A function can be perfectly continuous at a point while completely failing to be differentiable there!

- **Standard Counterexample:** $f(x) = |x|$ at $x = 0$. $\lim_{x\to 0}|x| = 0 = f(0)$ (Continuous). $Lf'(0) = \lim_{h\to 0} \frac{|0-h|-0}{-h} = \texttt{frac}\{h\}\{-h\} = -1$; $Rf'(0) = \lim_{h\to 0} \frac{|0+h|-0}{h} = \texttt{frac}\{h\}\{h\} = +1$. Since $Lf'(0) \neq Rf'(0)$, $f(x) = |x|$ has a **sharp corner (kink)** at the origin and is non-differentiable there!

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Differentiation Techniques & Core Theorems

1. The Differentiation Arsenal

- **Chain Rule:** If $y = f(u)$ and $u = g(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$.
- **Implicit Differentiation:** When $F(x, y) = 0$ defines y implicitly: differentiate each term with respect to x, applying the chain rule to y-terms ($\frac{d}{dx}[y^n] = ny^{n-1}\frac{dy}{dx}$), then algebraically isolate $\frac{dy}{dx}$.

2. Derivatives of Inverse Trigonometric Functions

Function $y = f(x)$	Derivative $\frac{dy}{dx}$	Valid Domain
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	$x \in (-1, 1)$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$	$x \in (-1, 1)$
$\tan^{-1} x$	$\frac{1}{1+x^2}$	$x \in \mathbb{R}$
$\cot^{-1} x$	$-\frac{1}{1+x^2}$	$x \in \mathbb{R}$
$\sec^{-1} x$	$\frac{1}{\lvert x\rvert \sqrt{x^2-1}}$	$\lvert x\rvert > 1$
$\csc^{-1} x$	$-\frac{1}{\lvert x\rvert \sqrt{x^2-1}}$	$\lvert x\rvert > 1$

3. Logarithmic & Parametric Differentiation

- **Logarithmic Differentiation:** Used when functions have variable exponents $y = [u(x)]^{v(x)}$ or multi-factor products:
  1. Take natural logarithm: $\ln y = v(x) \ln u(x)$.
  2. Differentiate implicitly: $\frac{1}{y}\frac{dy}{dx} = v'(x)\ln u(x) + v(x)\frac{u'(x)}{u(x)}$.
  3. Multiply by y: $\frac{dy}{dx} = [u(x)]^{v(x)}\left[v'(x)\ln u(x) + \frac{v(x)u'(x)}{u(x)}\right]$.
- **Parametric Differentiation:** If $x = f(t)$ and $y = g(t)$, then:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{g'(t)}{f'(t)} \quad \text{(provided } f'(t) \neq 0\text{)}$$

The High-Yield Second-Order Parametric Trap

To find $\frac{d^2y}{dx^2}$ for parametric curves $x = f(t), y = g(t)$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{dy}{dx}\right] = \frac{d}{dt}\left[\frac{dy}{dx}\right] \cdot \underbrace{\frac{dt}{dx}}_{\text{CRUCIAL FACTOR!}} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}$$

MASSIVE NTA TRAP WARNING

Never calculate $\frac{d^2y}{dx^2}$ by simply dividing $\frac{d^2\frac{y}{d}t^2}{d^2\frac{x}{d}t^2}$! This produces a catastrophic error. You MUST differentiate $\frac{dy}{dx}$ with respect to t and then multiply by $\frac{dt}{dx} = \frac{1}{d\frac{x}{d}t}$.

4. Rolle's Theorem & Lagrange's Mean Value Theorem (LMVT)

- **Rolle's Theorem:** If f is:
  1. Continuous on $[a, b]$,
  2. Differentiable on (a, b), and
  3. $f(a) = f(b)$,

  then there exists at least one $c \in (a, b)$ such that $f'(c) = 0$ (horizontal tangent).
- **Lagrange's Mean Value Theorem (LMVT):** If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists at least one $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Geometrically, the tangent at $x = c$ is parallel to the secant line connecting $(a, f(a))$ and $(b, f(b))$.

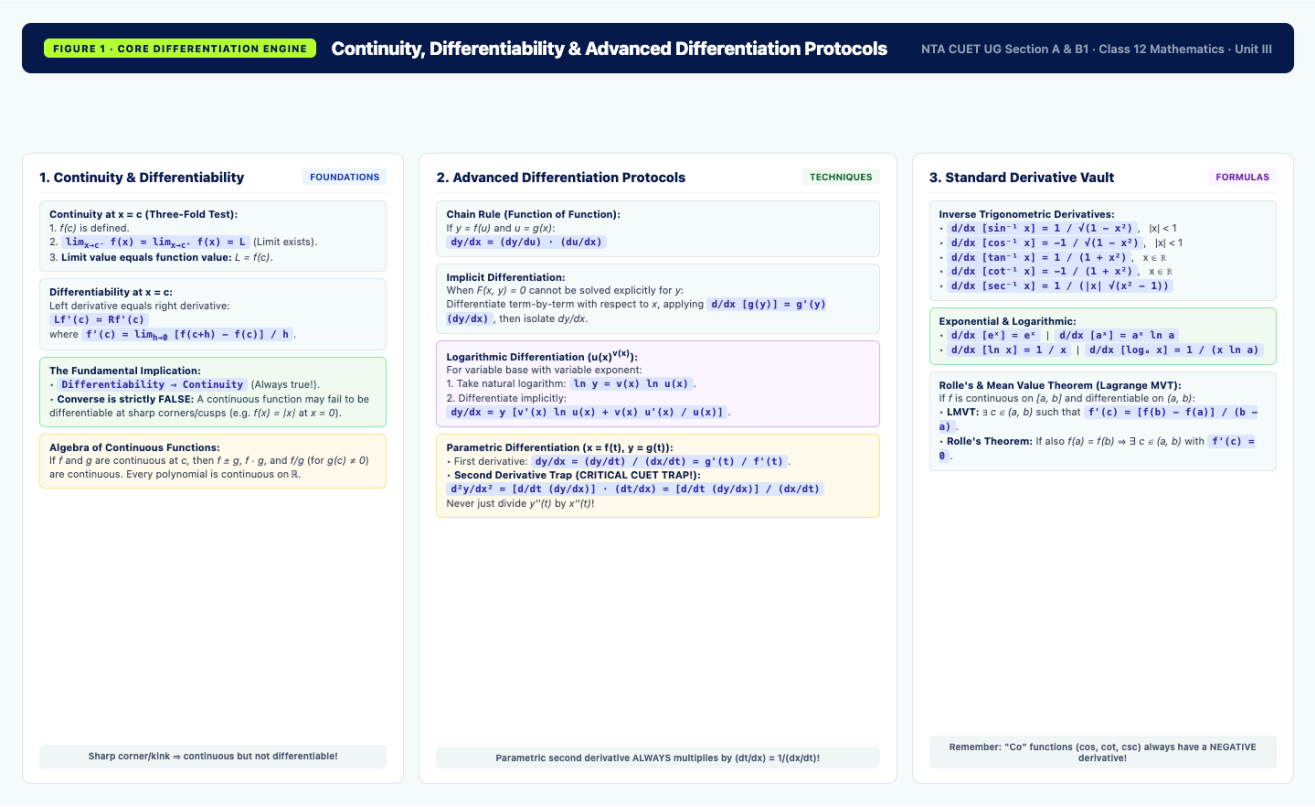


Figure 1: Figure 1: Core differentiation engine — Three-fold continuity checklist, differentiability vs continuity implication, logarithmic protocols, and the parametric second-derivative chain multiplier.

Key Analytical Takeaways from Figure 1:

1. **Continuity vs Differentiability Hierarchy:** Differentiability is a strictly stronger condition than continuity. Continuous functions can possess sharp corners ($|x|$ at 0), cusps ($x^{\frac{2}{3}}$ at 0), or vertical tangents ($x^{\frac{1}{3}}$ at 0) where derivatives fail to exist.
2. **Logarithmic Differentiation Requirement:** Whenever a function has the algebraic structure $[u(x)]^{v(x)}$ (variable base raised to variable power), direct power rules ($\frac{d}{dx}[x^n] = nx^{n-1}$) or exponential rules ($\frac{d}{dx}[a^x] = a^x \ln a$) do NOT apply. Natural logarithms must be taken first.
3. **Second-Order Parametric Execution:** When differentiating $\frac{dy}{dx}$ a second time, remember that the differentiation operator is with respect to x, while the expression is in terms of parameter t. The chain rule necessitates multiplying by $\frac{dt}{dx} = \frac{1}{f'(t)}$.

Rate of Change & Tangents/Normals

1. Rate of Change of Quantities

If a quantity y depends on another quantity x according to $y = f(x)$, then $\frac{dy}{dx}$ represents the **instantaneous rate of change of y with respect to x**. When both quantities vary with time t:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

- **Standard Geometric Models:**
  1. **Spherical Balloon (Radius r):**
  $$V = \frac{4}{3}\pi r^3 \Longrightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$
  $$S = 4\pi r^2 \Longrightarrow \frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$
  2. **Circular Wave / Ripple (Radius r):**
  $$A = \pi r^2 \Longrightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$
  3. **Sliding Ladder (Length L against wall):**
  $$x^2 + y^2 = L^2 \Longrightarrow 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Longrightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$$
  (The negative sign reflects that the height y is decreasing while base x is increasing).
  4. **Inverted Cone (Semi-vertical angle α):** Radius and height are linked by $r = h\tan\alpha$. Volume becomes $V = \frac{1}{3}\pi(h\tan\alpha)^2 h = \frac{1}{3}\pi\tan^2\alpha \cdot h^3$, allowing single-variable rate differentiation.

2. Slope & Equations of Tangents and Normals

Let $y = f(x)$ be a continuous curve. The **slope of the tangent** to the curve at point $P(x_0, y_0)$ is:

$$m = \left[\frac{dy}{dx}\right]_{(x_0,y_0)} = f'(x_0)$$

- **Equation of Tangent Line:**
$$y - y_0 = m(x - x_0)$$
- **Equation of Normal Line:** The normal line is perpendicular to the tangent line at $P(x_0, y_0)$. Its slope is $m_N = -\frac{1}{m}$ (for $m \neq 0$).
$$y - y_0 = -\frac{1}{m}(x - x_0)$$

Special Orientations of Tangents

- **Parallel to x-axis (Horizontal Tangent):** Angle $\theta = 0 \Longrightarrow m = \tan 0 = 0 \Longrightarrow \frac{dy}{dx} = 0$. Equation of horizontal tangent: $y = y_0$.
- **Perpendicular to x-axis (Vertical Tangent):** Angle $\theta = \frac{\pi}{2} \Longrightarrow m \to \infty \Longrightarrow \frac{dx}{dy} = 0$. Equation of vertical tangent: $x = x_0$.
- **Parallel to Line $y = m_0x + c$:** Set $\frac{dy}{dx} = m_0$ to find tangency coordinates (x_0, y_0).

Angle of Intersection of Two Curves

Let two curves $y = f_1(x)$ and $y = f_2(x)$ intersect at point (x_0, y_0) with tangent slopes m_1 and m_2. The acute angle of intersection θ is:

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

- **Orthogonal Curves:** Two curves intersect orthogonally (at right angles) if and only if:
$$m_1 \cdot m_2 = -1$$

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Monotonicity: Increasing & Decreasing Functions

1. Theoretical Definitions

Let f be a function defined on an interval I:

- **Strictly Increasing on I:** For all $x_1, x_2 \in I$,
$$x_1 < x_2 \implies f(x_1) < f(x_2)$$
- **Increasing (Non-decreasing) on I:** For all $x_1, x_2 \in I$,
$$x_1 < x_2 \implies f(x_1) \leq f(x_2)$$
- **Strictly Decreasing on I:** For all $x_1, x_2 \in I$,
$$x_1 < x_2 \implies f(x_1) > f(x_2)$$
- **Decreasing (Non-increasing) on I:** For all $x_1, x_2 \in I$,
$$x_1 < x_2 \implies f(x_1) \geq f(x_2)$$

2. Derivative Criteria for Monotonicity

Let f be continuous on $[a, b]$ and differentiable on (a, b):

1. f is **strictly increasing** on $[a, b]$ if $f'(x) > 0$ for all $x \in (a, b)$.
2. f is **strictly decreasing** on $[a, b]$ if $f'(x) < 0$ for all $x \in (a, b)$.
3. f is a **constant function** on $[a, b]$ if $f'(x) = 0$ for all $x \in (a, b)$.

The Discrete Zeroes Exception (Crucial Concept)

Even if $f'(x) = 0$ at isolated discrete points in (a, b), the function remains **strictly increasing** provided $f'(x) > 0$ everywhere else.

- **Classic NCERT Example:** $f(x) = x^3$. $f'(x) = 3x^2 \geq 0$. $f'(0) = 0$ only at the single isolated point $x = 0$. For any $x_1 < x_2$, $x_1^3 < x_2^3$ strictly. Thus, $f(x) = x^3$ is **strictly increasing on $\mathbb{R}$!**

3. Systematic Algorithm for Finding Monotonic Intervals

To determine the intervals where a differentiable function $f(x)$ is increasing or decreasing:

1. **Compute First Derivative:** Find $f'(x)$.
2. **Find Critical / Stationary Points:** Set $f'(x) = 0$ and find all real roots. Also include any points where $f'(x)$ fails to exist.
3. **Partition the Domain:** Plot these critical points on the real number line to divide the domain into disjoint sub-intervals.
4. **Sign Scheme (Wavy Curve Method):** Test the algebraic sign of $f'(x)$ in each sub-interval:
   - If $f'(x) > 0$ on $(c_i, c_{i+1}) \implies f(x)$ is strictly increasing on (c_i, c_{i+1}).
   - If $f'(x) < 0$ on $(c_i, c_{i+1}) \implies f(x)$ is strictly decreasing on (c_i, c_{i+1}).

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Maxima, Minima & Optimization Engine

1. Local Extrema & Critical Points

Let f be a real-valued function defined on domain D:

- **Local Maximum:** f has a local maximum at $x = c$ if there exists an open interval $(c - \delta, c + \delta)$ around c such that $f(c) \geq f(x)$ for all x in that neighborhood.
- **Local Minimum:** f has a local minimum at $x = c$ if $f(c) \leq f(x)$ for all x in that neighborhood.
- **Critical Point:** A point c in the domain of f where either:

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ fails to exist}$$

Every local extremum of an interior point occurs at a critical point (**Fermat's Theorem**).

2. The First Derivative Test

Let f be continuous at critical point c:

1. **Local Maximum:** As x increases through c, if $f'(x)$ changes sign from **positive to negative** $(+ \to -)$, then $x = c$ is a point of local maximum.
2. **Local Minimum:** As x increases through c, if $f'(x)$ changes sign from **negative to positive** $(- \to +)$, then $x = c$ is a point of local minimum.
3. **Point of Inflection:** If $f'(x)$ **does not change sign** as x increases through c, then c is neither a local maximum nor a local minimum (e.g. $f(x) = x^3$ at $x = 0$).

3. The Second Derivative Test

Let f be twice differentiable at c such that $f'(c) = 0$:

1. If $f''(c) < 0$, then f has a **local maximum** at c (concave downward).
2. If $f''(c) > 0$, then f has a **local minimum** at c (concave upward).
3. If $f''(c) = 0$, the second derivative test **fails (inconclusive)**! You must immediately revert to the First Derivative Test.

4. Absolute (Global) Extrema on Closed Intervals $[a, b]$

ABSOLUTE EXTREMA ALGORITHM

To find the absolute maximum and absolute minimum of a continuous function on $[a, b]$:

1. Find all critical points $c_1, c_2, \ldots \in (a, b)$ where $f'(c) = 0$ or f' does not exist.
2. Evaluate the function values at all critical points: $f(c_1), f(c_2), \ldots$
3. Evaluate the function values at the boundary endpoints: $f(a)$ and $f(b)$.
4. Compare all values:

$$\text{Absolute Maximum} = \max\{f(a), f(b), f(c_1), f(c_2), \ldots\}$$
$$\text{Absolute Minimum} = \min\{f(a), f(b), f(c_1), f(c_2), \ldots\}$$

Warning: Never omit the endpoints a and b! The absolute extremum frequently occurs at a boundary where $f'(x) \neq 0$.

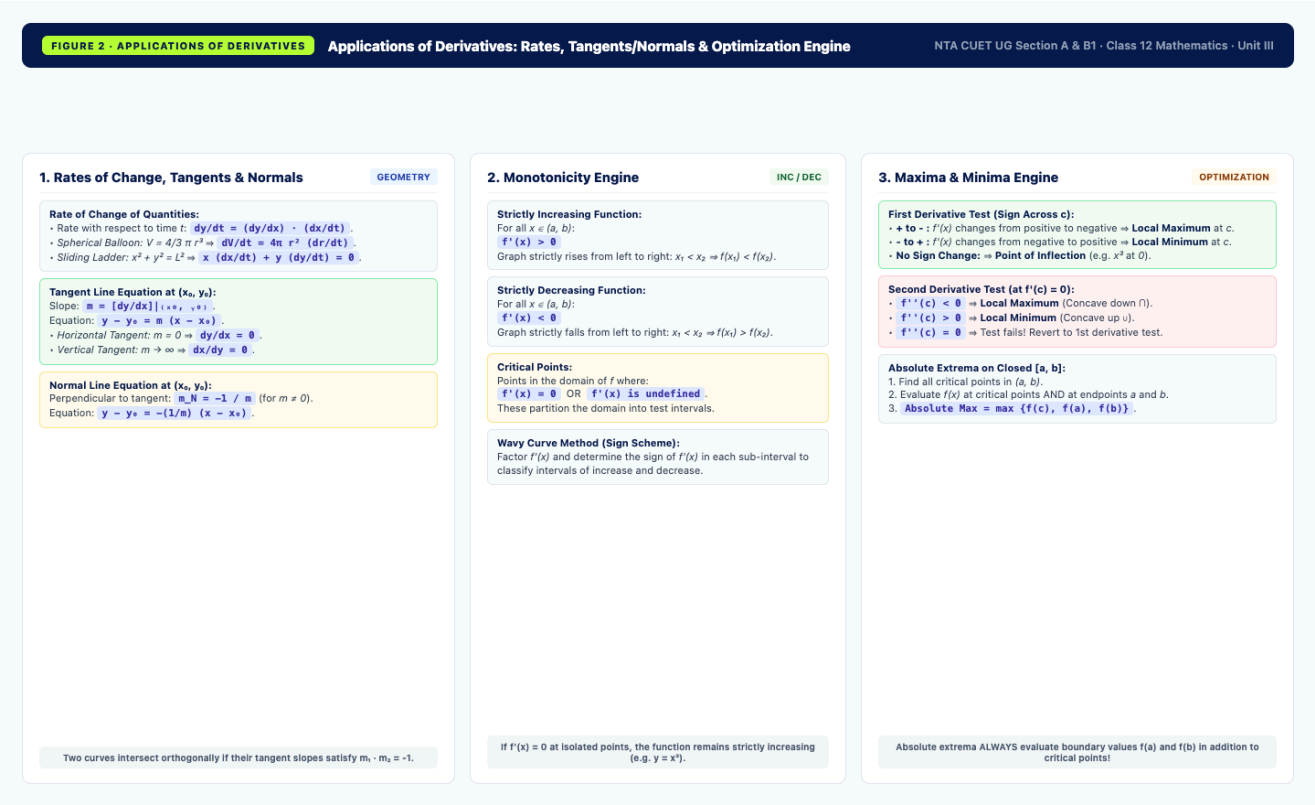


Figure 2: Figure 2: Applications of derivatives engine — Geometric rate measure equations, tangent/normal slope relationships, monotonicity intervals, and first/second derivative optimization tests.

Key Analytical Takeaways from Figure 2:

1. **Rate Measure Chain Links:** Time derivatives of geometric quantities require linking through spatial derivatives. When solving balloon or ladder problems, always express the volume/area in a single variable before taking $\frac{d}{dt}$.
2. **Tangent vs Normal Slopes:** Tangent slope is $m = f'(x_0)$, while normal slope is $m_N = -\frac{1}{f'}(x_0)$. For vertical tangents ($m \to \infty$), the normal is horizontal ($m_N = 0$).
3. **Second Derivative Sign Rule:** A negative second derivative ($f'' < 0$) indicates local **maximum** (concave down), while a positive second derivative ($f'' > 0$) indicates local **minimum** (concave up). Many students mistakenly associate "negative" with minimum.

Applied Optimization & Linear Approximations

1. Master Applied Optimization Protocol

Real-life optimization word problems follow a strict 4-step blueprint:

1. **Identify Objective & Constraints:** Determine the quantity to be maximized or minimized (Q) and write down geometric/physical constraints.
2. **Single-Variable Reduction:** Use constraint equations to eliminate extra variables so that $Q = f(x)$ depends on **strictly one independent variable**.
3. **Critical Point Isolation:** Compute $f'(x)$, set $f'(x) = 0$, and solve for candidate x.
4. **Extremum Verification:** Confirm with $f''(x)$ (< 0 for max, > 0 for min).

Standard NCERT Optimization Archetypes

- **Open Rectangular Box:** Cutting squares of side x from corners of a sheet of dimensions $a \times b$ and folding up:
$$V(x) = x(a - 2x)(b - 2x)$$
For a square sheet ($a = b$): $V(x) = x(a - 2x)^2 \implies x = \frac{a}{6}$ yields maximum volume.
- **Cylinder Inscribed in Sphere (Radius R):** Let cylinder have radius r and height h. Constraint: $r^2 + \left(\frac{h}{2}\right)^2 = R^2 \implies r^2 = R^2 - \frac{h^2}{4}$. Volume $V = \pi r^2 h = \pi\left(R^2 h - \frac{h^3}{4}\right)$. $\frac{dV}{dh} = \pi\left(R^2 - 3\frac{h^2}{4}\right) = 0 \implies h = \frac{2R}{\sqrt{3}}$. Cylinder of maximum volume has height $h = \frac{2R}{\sqrt{3}}$ and radius $r = \sqrt{\frac{2}{3}}R$.
- **Cone Inscribed in Sphere (Radius R):** Maximum volume cone inscribed in a sphere of radius R has altitude $h = \frac{4}{3}R$, and its volume is $\frac{8}{27}$ of the sphere's volume.
- **Wire Cutting Problem (Circle & Square):** A wire of length L is cut into two pieces to form a square (perimeter $4s$) and a circle (circumference $2\pi r$). The combined area $A = s^2 + \pi r^2$ is **minimum** when:
$$2r = s \quad \text{(Diameter of circle equals side of square!)}$$

2. Differentials & Linear Approximation

Let $y = f(x)$. If x changes by a small amount $\Delta x = dx$, the corresponding change in y is approximated by the differential dy:
$$\Delta y \approx dy = f'(x)dx = f'(x)\Delta x$$

- **Linear Approximation Formula:**
$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x$$
- **Error Measurements:**
  - Absolute Error: Δx
  - Relative Error: $\frac{\Delta x}{x}$
  - Percentage Error: $\frac{\Delta x}{x} \times 100\%$

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Key Facts & Terms Cheatsheet

MASTER DIAGNOSTIC & CONCEPTUAL CHEATSHEET — DIFFERENTIAL CALCULUS

Key Calculus Topic	Core NCERT Mechanism / Definition	Exam Identifier
Continuity Test	$\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = f(c)$. Graph is unbroken.	LHL = RHL = f(c)
Diff Implies Cont	Differentiability $\Longrightarrow$ Continuity. Converse is strictly false ($\lvert x\rvert$ at 0).	Smooth $\Longrightarrow$ Continuous
Corner Point	Point where $Lf'(c) \neq Rf'(c)$. Continuous, but not differentiable.	Kink / Modulus
Chain Rule	$d\frac{y}{d}x = \left(d\frac{y}{d}u\right)\cdot\left(d\frac{u}{d}x\right)$. Used for nested functions.	Composite Functions
Implicit Diff	Differentiate term-by-term; $\frac{d}{dx}[g(y)] = g'(y)\left(d\frac{y}{d}x\right)$. Solve for y'.	F(x, y) = 0
Logarithmic Diff	For $y = u(x)^{v(x)}$, take $\ln y = v\ln u$ before differentiating.	Tower of Functions
Parametric First Deriv	$d\frac{y}{d}x = \frac{d\frac{y}{d}t}{d\frac{x}{d}t} = g'\frac{t}{f'}(t)$.	Ratio of Rates
Parametric Second Deriv	$d^2\frac{y}{d}x^2 = \frac{\left[\frac{d}{dt}\left(d\frac{y}{d}x\right)\right]}{d\frac{x}{d}t}$. Never $y''\frac{t}{x''}(t)$!	Multiply by dt/dx
Inverse Trig Deriv	$\frac{d}{dx}[\sin^{-1}x] = \frac{1}{\sqrt{1-x^2}}$; $\frac{d}{dx}[\tan^{-1}x] = \frac{1}{1+x^2}$.	Inverse Trig Vault
Rate of Change	$d\frac{y}{d}t = \left(d\frac{y}{d}x\right)\cdot\left(d\frac{x}{d}t\right)$. Geometric volume/surface rates.	Time Derivative
Tangent Slope	$m = \left[d\frac{y}{d}x\right]_{(x_0,y_0)}$. Equation: $y - y_0 = m(x - x_0)$.	Tangent Slope
Normal Slope	$m_N = -\frac{1}{m}$. Equation: $y - y_0 = \left(-\frac{1}{m}\right)(x - x_0)$.	Negative Reciprocal
Horizontal Tangent	$m = 0 \Longrightarrow d\frac{y}{d}x = 0$. Tangent is parallel to x-axis.	dy/dx = 0
Vertical Tangent	$m \to \infty \Longrightarrow d\frac{x}{d}y = 0$. Tangent is perpendicular to x-axis.	dx/dy = 0
Orthogonal Curves	Curves intersect at right angles $\Longleftrightarrow m_1 \cdot m_2 = -1$.	$m_1\ m_2 = -1$
Strictly Increasing	$f'(x) > 0$ on (a, b). Function rises continuously from left to right.	f'(x) > 0
Strictly Decreasing	$f'(x) < 0$ on (a, b). Function falls continuously from left to right.	f'(x) < 0
Critical Point	Point in domain where $f'(x) = 0$ or $f'(x)$ is undefined.	Stationary / Cusp
First Deriv Test	$+ \to -$ is Local Max; $- \to +$ is Local Min; no sign change is Inflection.	Sign Change across c
Second Deriv Test	$f'(c) = 0$ and $f''(c) < 0 \Longrightarrow$ Local Max; $f''(c) > 0 \Longrightarrow$ Local Min.	Concavity Test
Absolute Extrema	Evaluate $f(x)$ at critical points AND endpoints a, b on $[a, b]$.	Include Boundaries!
Linear Approximation	$f(x + \Delta x) \approx f(x) + f'(x)\Delta x$. Relative error $\Delta\frac{x}{x}$.	Differentials

Common Errors and High-Yield CUET Traps

COMMON STUDENT MISTAKES & DIAGNOSTIC TRAPS — DIFFERENTIAL CALCULUS

Common Student Mistake	Scientifically Verified Reality	The Exact Trap Mechanism
Parametric Second Derivative Division	$d^2\frac{y}{d}x^2 = \left[\frac{d}{dt}\left(d\frac{y}{d}x\right)\right] \cdot \left(d\frac{t}{d}x\right)$. It is NOT equal to $\frac{d^2\frac{y}{d}t^2}{d^2\frac{x}{d}t^2}$.	Students differentiate y twice and x twice and divide them, omitting the essential $\left(d\frac{t}{d}x\right)$ chain factor.
Neglecting Endpoints in Absolute Extrema	On a closed interval $[a, b]$, absolute extrema often occur at the boundaries $x = a$ or $x = b$, where $f'(x) \neq 0$.	Students find critical points where $f'(x) = 0$ and ignore $f(a)$ and $f(b)$, missing the true maximum value.
Assuming Continuous Implies Differentiable	Differentiability requires smoothness. Continuous functions can have sharp corners or vertical tangents where derivatives fail.	Questions ask if $\lvert x - 1\rvert$ is differentiable at $x = 1$; students see it is continuous and incorrectly select differentiable.
Inverting Second Derivative Test Signs	$f''(c) < 0$ corresponds to a **Local Maximum** (concave down), whereas $f''(c) > 0$ corresponds to a **Local Minimum** (concave up).	Students associate negative with "minimum" and positive with "maximum", reversing local extrema classifications.
Using Power Rule on $[u(x)]^{v(x)}$	Functions with variable exponents require **logarithmic differentiation**. Direct power or exponential rules are invalid.	Students write $\frac{d}{dx}[x^x] = x \cdot x^{x-1} = x^x$, missing the complete term $x^{x(1+\ln x)}$.
Forgetting Inconclusive $f''(c) = 0$	If $f'(c) = 0$ and $f''(c) = 0$, the second derivative test yields **no conclusion**. Must use the First Derivative Test.	Students declare $f''(c) = 0$ as an automatic point of inflection without checking higher derivatives or sign changes.
Wrong Sign in Sliding Ladder Rate	Since $x^2 + y^2 = L^2$, $x\left(d\frac{x}{d}t\right) + y\left(d\frac{y}{d}t\right) = 0$. As x increases, y decreases, so $d\frac{y}{d}t$ must be negative.	Students report positive rates of descent for ladder tops, leading to sign errors in MCQ answer keys.
Applying Rolle's Theorem Without $f(a) = f(b)$	Rolle's Theorem requires THREE conditions: continuity on $[a, b]$, differentiability on (a, b), AND $f(a) = f(b)$.	Students apply $f'(c) = 0$ to functions where $f(a) \neq f(b)$, where only Lagrange's MVT is valid.

OurMocks

The Whole Chapter on One Page

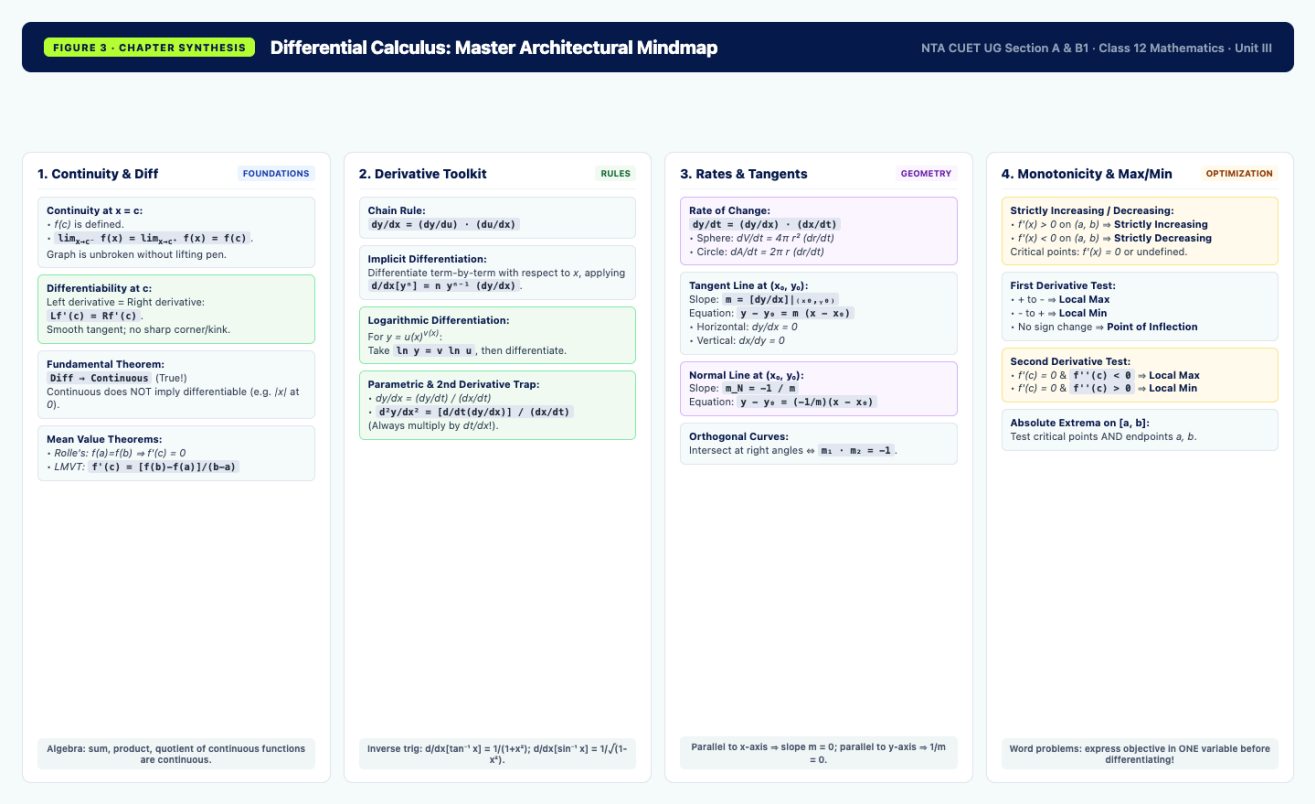


Figure 3: Figure 3: Master synthesis mindmap — integrated architecture of Continuity, Differentiability, Advanced Derivative Toolkit, Tangents/Normals, and Optimization Engine.

Practice Questions — Part 1 (Q1 to Q5)

Q1. If the function $f(x) = \begin{cases} \frac{k\cos x}{\pi - 2x} & \text{if } x \neq \frac{\pi}{2} \\ 3 & \text{if } x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$, then the value of k is:

(A) 3 (B) 6
(C) 12 (D) $\frac{3}{2}$

Sol. Q1. For continuity at $x = \frac{\pi}{2}$, we must have $\lim_{x\to\frac{\pi}{2}} f(x) = f\left(\frac{\pi}{2}\right) = 3$. Evaluate the limit by substitution $x = \frac{\pi}{2} + h$ as $h \to 0$:

$$\lim_{x\to\frac{\pi}{2}} \frac{k\cos x}{\pi - 2x} = \lim_{h\to 0} \frac{k\cos\left(\frac{\pi}{2} + h\right)}{\pi - 2\left(\frac{\pi}{2} + h\right)} = \lim_{h\to 0} \frac{-k\sin h}{-2h} = \frac{k}{2}\lim_{h\to 0}\frac{\sin h}{h} = \frac{k}{2}$$

Equating limit value to $f\left(\frac{\pi}{2}\right)$:

$$\frac{k}{2} = 3 \Longrightarrow k = 6$$

Answer: (B)

Q2. Let $f(x) = |x-2| + |x-3|$ for all $x \in \mathbb{R}$. The set of points where $f(x)$ is NOT differentiable is:

(A) $\{2\}$ (B) $\{3\}$
(C) $\{2, 3\}$ (D) $\mathbb{R}$ (not differentiable anywhere)

Sol. Q2. The modulus function $|x - c|$ is continuous everywhere on $\mathbb{R}$, but fails to be differentiable at the point where its argument vanishes ($x = c$) due to a sharp corner. Here, $f(x)$ is the sum of two modulus functions with corner points at $x = 2$ and $x = 3$. Evaluating left and right derivatives:

- At $x = 2$: $Lf'(2) = -2$, $Rf'(2) = 0 \Longrightarrow Lf'(2) \neq Rf'(2)$.
- At $x = 3$: $Lf'(3) = 0$, $Rf'(3) = 2 \Longrightarrow Lf'(3) \neq Rf'(3)$.

Hence, f is non-differentiable at exactly the two points $\{2, 3\}$. **Answer: (C)**

Q3. The derivative of $y = \tan^{-1}\left(\frac{3x - x^3}{1 - 3x^2}\right)$ with respect to x for $-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$ is equal to:

(A) $\frac{1}{1+x^2}$ (B) $\frac{3}{1+x^2}$
(C) $\frac{3}{1+9x^2}$ (D) $\frac{1}{1+9x^2}$

Sol. Q3. Substitute $x = \tan\theta \Longrightarrow \theta = \tan^{-1} x$. For $-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$, $-\frac{\pi}{6} < \theta < \frac{\pi}{6} \Longrightarrow -\frac{\pi}{2} < 3\theta < \frac{\pi}{2}$. The expression simplifies via the triple-angle identity:

$$\frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} = \tan(3\theta)$$

Therefore, $y = \tan^{-1}(\tan(3\theta)) = 3\theta = 3\tan^{-1} x$. Differentiating with respect to x:

$$\frac{dy}{dx} = 3 \cdot \frac{d}{dx}[\tan^{-1} x] = \frac{3}{1 + x^2}$$

Answer: (B)

Q4. If $x = a\cos^3\theta$ and $y = a\sin^3\theta$, then the value of $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{4}$ is:

(A) $\frac{4\sqrt{2}}{3a}$ (B) $-\frac{4\sqrt{2}}{3a}$
(C) $\frac{1}{3a}$ (D) $\frac{2\sqrt{2}}{a}$

Sol. Q4. Differentiating parametrically with respect to θ:

$$\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta, \quad \frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$$

$$\frac{dy}{dx} = \frac{d\frac{y}{d}\theta}{d\frac{x}{d}\theta} = \frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta} = -\tan\theta$$

Now compute the second derivative using the parametric chain rule:

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}[-\tan\theta] \cdot \frac{d\theta}{dx} = (-\sec^2\theta) \cdot \frac{1}{-3a\cos^2\theta\sin\theta} = \frac{\sec^4\theta}{3a\sin\theta}$$

Substitute $\theta = \frac{\pi}{4}$:

$$\sec\left(\frac{\pi}{4}\right) = \sqrt{2} \Longrightarrow \sec^4\left(\frac{\pi}{4}\right) = 4; \quad \sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\frac{d^2y}{dx^2} = \frac{4}{3a\left(\frac{1}{\sqrt{2}}\right)} = \frac{4\sqrt{2}}{3a}$$

Answer: (A)

Q5. If $y = x^x$, then the value of the derivative $\frac{dy}{dx}$ at $x = 1$ is equal to:

(A) 0 (B) 1
(C) e (D) $1 + \ln 2$

Sol. Q5. Since the base and exponent both contain the variable x, we apply logarithmic differentiation:

$$\ln y = x\ln x$$

Differentiating implicitly with respect to x:

$$\frac{1}{y}\frac{dy}{dx} = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$$

$$\frac{dy}{dx} = y(1 + \ln x) = x^x(1 + \ln x)$$

At $x = 1$:

$$\frac{dy}{dx} = 1^1(1 + \ln 1) = 1(1 + 0) = 1$$

Answer: (B)

Practice Questions — Part 2 (Q6 to Q10)

Q6. A spherical balloon is inflated so that its volume increases at the rate of $900\text{cm}^3/\text{ sec}$. The rate of increase of its surface area when its radius is 15 cm is:

(A) $120\text{cm}^2/\text{ sec}$ **(B)** $240\text{cm}^2/\text{ sec}$
(C) $60\text{cm}^2/\text{ sec}$ **(D)** $180\text{cm}^2/\text{ sec}$

Sol. Q6. Volume of sphere: $V = \frac{4}{3}\pi r^3 \Longrightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. Given $\frac{dV}{dt} = 900\text{cm}^3/\text{ s}$ and $r = 15$ cm:

$$900 = 4\pi(15)^2 \frac{dr}{dt} = 900\pi \frac{dr}{dt} \Longrightarrow \frac{dr}{dt} = \frac{1}{\pi} \text{ cm/s}$$

Surface area of sphere: $S = 4\pi r^2 \Longrightarrow \frac{dS}{dt} = 8\pi r \frac{dr}{dt}$. Substituting $r = 15$ and $\frac{dr}{dt} = \frac{1}{\pi}$:

$$\frac{dS}{dt} = 8\pi(15)\left(\frac{1}{\pi}\right) = 120\text{cm}^2/\text{ sec}$$

Answer: (A)

Q7. The point on the curve $y = x^3 - 11x + 5$ at which the tangent line is $y = x - 11$ is:

(A) $(2, -9)$ **(B)** $(-2, 19)$
(C) $(3, -1)$ **(D)** $(2, 9)$

Sol. Q7. Slope of the given line $y = x - 11$ is $m = 1$. Slope of tangent to the curve is:

$$\frac{dy}{dx} = 3x^2 - 11 = 1 \Longrightarrow 3x^2 = 12 \Longrightarrow x^2 = 4 \Longrightarrow x = \pm 2$$

Test both points on the curve:

1. For $x = 2$: $y = (2)^3 - 11(2) + 5 = 8 - 22 + 5 = -9$. Point is $(2, -9)$. Equation of tangent: $y - (-9) = 1(x - 2) \Longrightarrow y + 9 = x - 2 \Longrightarrow y = x - 11$. (Matches!).
2. For $x = -2$: $y = (-2)^3 - 11(-2) + 5 = -8 + 22 + 5 = 19$. Point is $(-2, 19)$. Equation of tangent: $y - 19 = 1(x + 2) \Longrightarrow y = x + 21 \neq y = x - 11$.

Thus, the unique tangency point is $(2, -9)$. **Answer: (A)**

Q8. The curves $y^2 = 4x$ and $x^2 = 4y$ intersect at the point $(4, 4)$. The tangent of the angle between them at this point is:

(A) $\frac{3}{4}$ **(B)** $\frac{4}{3}$
(C) 1 **(D)** $\frac{1}{2}$

Sol. Q8. Find the slopes of both curves at $(4, 4)$:

1. Curve 1: $y^2 = 4x \Longrightarrow 2y\frac{dy}{dx} = 4 \Longrightarrow m_1 = \frac{4}{2y} = \frac{4}{2(4)} = \frac{1}{2}$.
2. Curve 2: $x^2 = 4y \Longrightarrow 2x = 4\frac{dy}{dx} \Longrightarrow m_2 = \frac{2x}{4} = \frac{2(4)}{4} = 2$.

The angle θ between the tangents is given by:

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{2 - \frac{1}{2}}{1 + \left(\frac{1}{2}\right)(2)}\right| = \texttt{frac}\left\{\frac{3}{2}\right\}\{2\} = \frac{3}{4}$$

Answer: (A)

Q9. The function $f(x) = 2x^3 - 9x^2 + 12x + 15$ is strictly increasing in the interval:

(A) $(1, 2)$ **(B)** $(-\infty, 1) \cup (2, \infty)$
(C) $(-1, 2)$ **(D)** $(-\infty, -2) \cup (-1, \infty)$

Sol. Q9. Differentiate $f(x)$ with respect to x:

$$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2)$$

For strictly increasing behavior, $f'(x) > 0$:

$$6(x - 1)(x - 2) > 0$$

Using the Wavy Curve Method, the roots are $x = 1$ and $x = 2$. $f'(x) > 0$ when $x < 1$ or $x > 2$. Hence, f is strictly increasing in $(-\infty, 1) \cup (2, \infty)$. **Answer: (B)**

Q10. The function $f(x) = \frac{\ln x}{x}$ is strictly decreasing in the interval:

(A) $(0, 1)$ **(B)** $(1, e)$
(C) (e, ∞) **(D)** $(0, e)$

Sol. Q10. Domain of $f(x)$ is $x \in (0, \infty)$. Differentiating via the quotient rule:

$$f'(x) = \texttt{frac}\left\{x \cdot \left(\frac{1}{x}\right) - (\ln x) \cdot 1\right\}\{x^2\} = \frac{1 - \ln x}{x^2}$$

For f to be strictly decreasing, $f'(x) < 0$:

$$\frac{1 - \ln x}{x^2} < 0 \Longrightarrow 1 - \ln x < 0 \quad (\text{since } x^2 > 0 \text{ for } x > 0)$$

$$\ln x > 1 \Longrightarrow x > e$$

Thus, $f(x)$ is strictly decreasing in (e, ∞). **Answer: (C)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. The maximum value of the function $f(x) = \left(\frac{1}{x}\right)^x$ for $x > 0$ is:

(A) e **(B)** $e^{\frac{1}{e}}$
(C) $\left(\frac{1}{e}\right)^e$ **(D)** 1

Sol. Q11. Let $y = \left(\frac{1}{x}\right)^x = x^{-x}$. Taking natural logarithms: $\ln y = -x \ln x$. Differentiating with respect to x: $\frac{1}{y}\frac{dy}{dx} = -\left(1 \cdot \ln x + x \cdot \frac{1}{x}\right) = -(\ln x + 1)$. Setting $\frac{dy}{dx} = 0 \Longrightarrow -(\ln x + 1) = 0 \Longrightarrow \ln x = -1 \Longrightarrow x = e^{-1} = \frac{1}{e}$. Testing concavity: $f''\left(\frac{1}{e}\right) < 0$, confirming a local maximum at $x = \frac{1}{e}$. The maximum value is:

$$f\left(\frac{1}{e}\right) = \left(\frac{1}{\frac{1}{e}}\right)^{\frac{1}{e}} = e^{\frac{1}{e}}$$

Answer: (B)

Q12. The absolute maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x + 1$ on the closed interval $[1, 5]$ is:

(A) 29 **(B)** 28
(C) 56 **(D)** 24

Sol. Q12. Find critical points by setting $f'(x) = 0$:

$$f'(x) = 6x^2 - 30x + 36 = 6(x^2 - 5x + 6) = 6(x-2)(x-3) = 0 \Longrightarrow x = 2, 3$$

Both critical points lie inside $(1, 5)$. Evaluate $f(x)$ at critical points and boundary endpoints $x = 1, 5$:

- $f(1) = 2(1) - 15(1) + 36(1) + 1 = 24$; $f(2) = 2(8) - 15(4) + 36(2) + 1 = 29$
- $f(3) = 2(27) - 15(9) + 36(3) + 1 = 28$; $f(5) = 2(125) - 15(25) + 36(5) + 1 = 56$

Comparing all values, the absolute maximum value is 56 (achieved at endpoint $x = 5$). **Answer: (C)**

Q13. A wire of length 28 m is cut into two pieces. One piece forms a square (side s) and the other forms a circle (radius r). The combined area is minimum when:

(A) $s = r$ **(B)** $s = 2r$
(C) $s = \pi r$ **(D)** $s = 4r$

Sol. Q13. Perimeter constraint: $4s + 2\pi r = 28 \Longrightarrow 2s + \pi r = 14 \Longrightarrow s = \frac{14 - \pi r}{2}$. Combined area: $A(r) = s^2 + \pi r^2 = \frac{(14-\pi r)^2}{4} + \pi r^2$. Differentiating with respect to r: $\frac{dA}{dr} = -\frac{\pi(14-\pi r)}{2} + 2\pi r = 0 \Longrightarrow -(14 - \pi r) + 4r = 0 \Longrightarrow r = \frac{14}{\pi+4}$. Then side of square $s = \frac{14-\pi r}{2} = \frac{28}{\pi+4}$. Comparing: $2r = 2\left(\frac{14}{\pi+4}\right) = \frac{28}{\pi+4} = s$. Thus, combined area is minimum when the side of the square equals the diameter of the circle ($s = 2r$). **Answer: (B)**

Q14. The value of c in Rolle's Theorem for $f(x) = x^3 - 4x$ on the interval $[-2, 2]$ is:

(A) 0 **(B)** $\pm\frac{2}{\sqrt{3}}$
(C) $\pm\frac{1}{\sqrt{3}}$ **(D)** ± 2

Sol. Q14. Check the conditions of Rolle's Theorem:

1. $f(x)$ is a polynomial, so continuous on $[-2, 2]$.
2. $f(x)$ is differentiable on $(-2, 2)$.
3. Boundary values: $f(-2) = (-2)^3 - 4(-2) = -8 + 8 = 0$; $f(2) = 2^3 - 4(2) = 8 - 8 = 0 \Longrightarrow f(-2) = f(2)$.

All conditions are met. By Rolle's Theorem, there exists $c \in (-2, 2)$ such that $f'(c) = 0$:

$$f'(c) = 3c^2 - 4 = 0 \Longrightarrow 3c^2 = 4 \Longrightarrow c^2 = \frac{4}{3} \Longrightarrow c = \pm\frac{2}{\sqrt{3}}$$

Both values lie strictly inside $(-2, 2)$ since $\frac{2}{\sqrt{3}} \approx 1.155 < 2$. **Answer: (B)**

Q15. Using differentials, the approximate value of $\sqrt{25.3}$ is equal to:

(A) 5.03 **(B)** 5.06
(C) 5.015 **(D)** 5.30

Sol. Q15. Let $f(x) = \sqrt{x}$. Choose $x = 25$ (nearest perfect square) and $\Delta x = 0.3$. Derivative: $f'(x) = \frac{1}{2\sqrt{x}}$. At $x = 25$:

$$f(25) = \sqrt{25} = 5, \quad f'(25) = \frac{1}{2\sqrt{25}} = \frac{1}{10} = 0.1$$

Using the differential approximation formula:

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x$$
$$\sqrt{25.3} \approx 5 + (0.1)(0.3) = 5 + 0.03 = 5.03$$

Answer: (A)